Lab09: Candy Mini Project

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Import Candy Data

```r
candy_file <- "candy-data.csv"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
```

```
One quarter      0    0          0          0.011          0.511    46.11650
Air Heads        0    0          0          0.906          0.511    52.34146
Almond Joy       0    1          0          0.465          0.767    50.34755
```

   Q1. How many different candy types are in this dataset?

There are 85 candy types.

```r
# Counts how many rows there are
nrow(candy)
```

```
[1] 85
```

   Q2. How many fruity candy types are in the dataset?

There are 38 fruity candy types.

```r
# Counts how many candy have fruit = 1
sum(candy$fruity)
```

```
[1] 38
```

```r
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy is Snickers and it's winpercent value is 76.67378.

```r
candy["Snickers", ]$winpercent
```

```
[1] 76.67378
```

> Q4. What is the winpercent value for "Kit Kat"?

The winpercent value for "Kit Kat" is 76.7686.

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

> Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

The winpercent value for Tootsie Roll is 49.6535.

```r
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```r
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85

Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent column looks to be on a different scale compared to the other columns. The majority of the other columns in the dataset have values ranging from 0 to 1 and winpercent ranges from 20 to 85.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

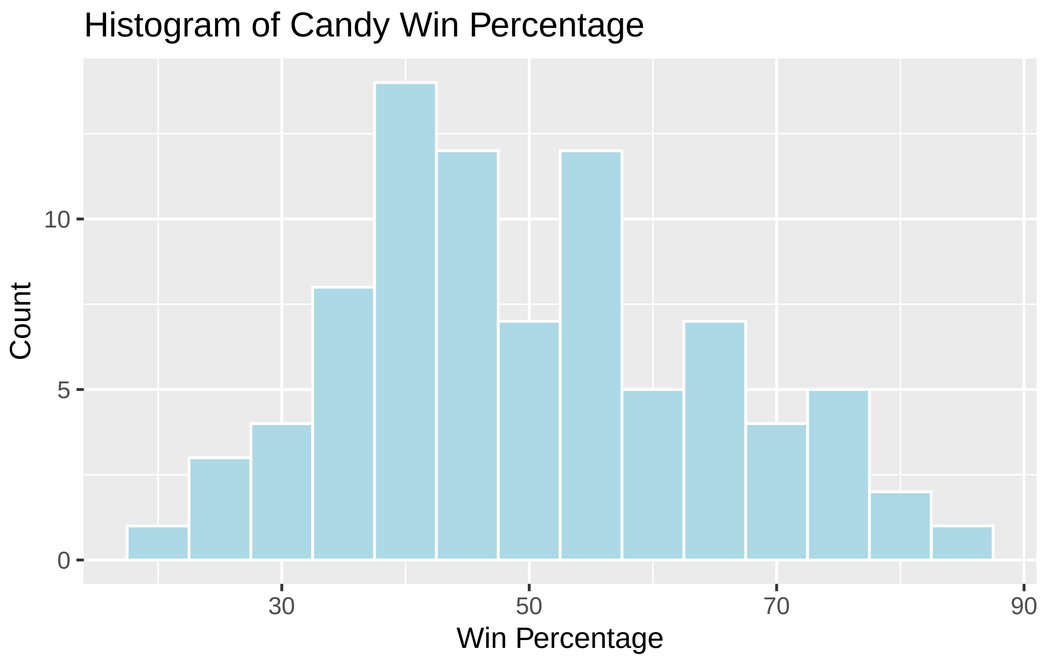
A 0 represents the candy is not a chocolate and a 1 represents that the candy is chocolate/has chocolate for the candy$chocolate column.

Exploratory Analysis

> Q8. Plot a histogram of winpercent values using both base R an ggplot2.

```
library(ggplot2)

ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 5, color = "white", fill = "lightblue") +
  labs(title = "Histogram of Candy Win Percentage",
       x = "Win Percentage",
       y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values does not look perfectly symmetrical.

Q10. Is the center of the distribution above or below 50%?

Based on the `skim()` output, the median for p50 for winpercent is 47.83. The center of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[as.logical((candy$chocolate))])
```

```
[1] 60.92153
```

```r
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

On average chocolate candy is ranked higher than fruit candy. This is because chocolate candy have a higher mean winpercent, 60.92, compared to the winpercent for fruit candy, 44.12.

> Q12. Is this difference statistically significant?

```r
t.test(candy$winpercent[as.logical(candy$chocolate)], candy$winpercent[as.logical(candy$frui
```

```

	Welch Two Sample t-test

data:  candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fi
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

The difference is statistically significant. The p-value is less than 0.05, which means chocolate candy have a significantly higher average winpercent then fruit candy.

Overall Candy Rankings

> Q13. What are the five least liked candy types in this set?

The least 5 liked candy types are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```r
library(dplyr)

candy |>
  arrange(winpercent) |>
  head(5)
```

```
                  chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                 0      1       0              0      0
Boston Baked Beans        0      0       0              1      0
Chiclets                  0      1       0              0      0
Super Bubble              0      1       0              0      0
Jawbusters                0      1       0              0      0
                  crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                        0    0   0        1        0.197        0.976
Boston Baked Beans               0    0   0        1        0.313        0.511
Chiclets                         0    0   0        1        0.046        0.325
Super Bubble                     0    0   0        0        0.162        0.116
Jawbusters                       0    1   0        1        0.093        0.511
                  winpercent
Nik L Nip           22.44534
Boston Baked Beans  23.41782
Chiclets            24.52499
Super Bubble        27.30386
Jawbusters          28.12744
```

Q14. What are the top 5 all time favorite candy types out of this set?

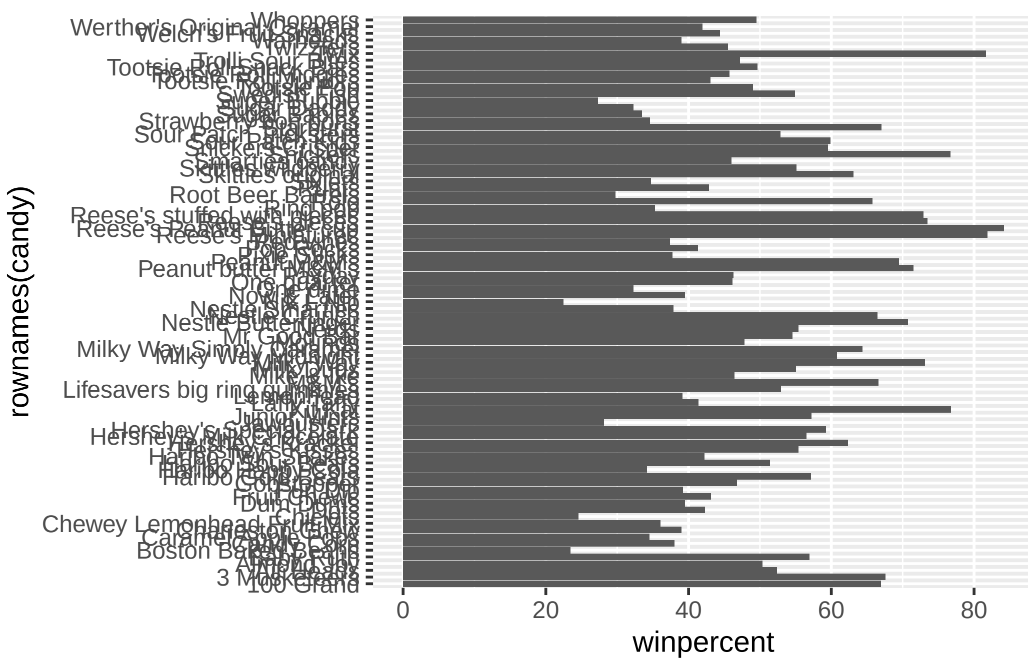
```r
candy |>
  arrange(desc(winpercent)) |>
  head(5)
```

```
                         chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup        1      0       0              1      0
Reese's Miniatures               1      0       0              1      0
Twix                             1      0       1              0      0
Kit Kat                          1      0       0              0      0
Snickers                         1      0       1              1      1
                         crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                         pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

The top 5 favorite candy are Reese's Peanut Butter Cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

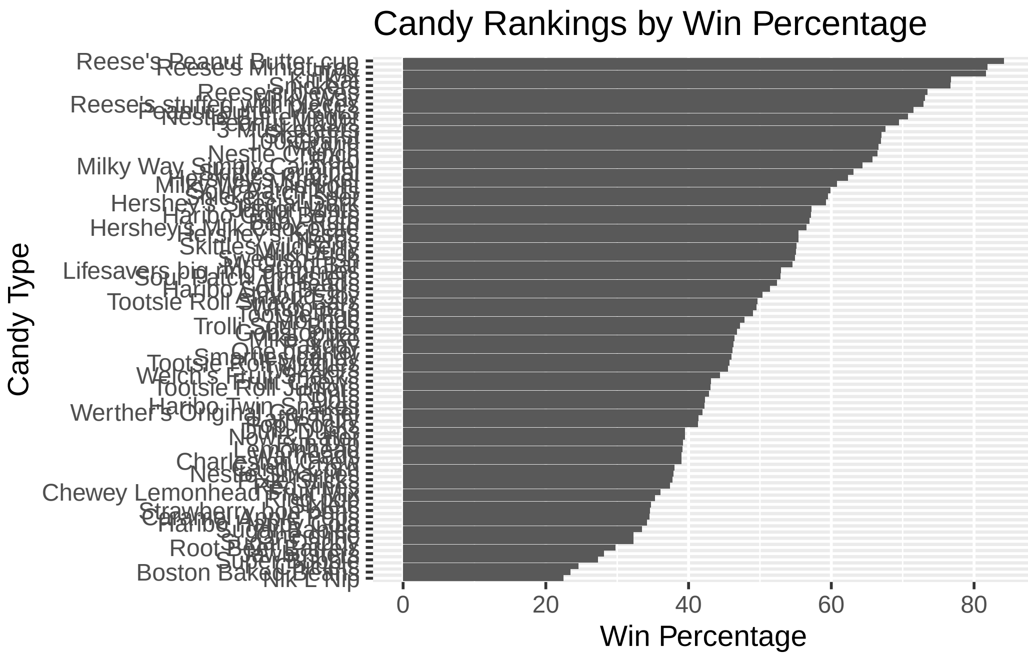
> Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



> Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

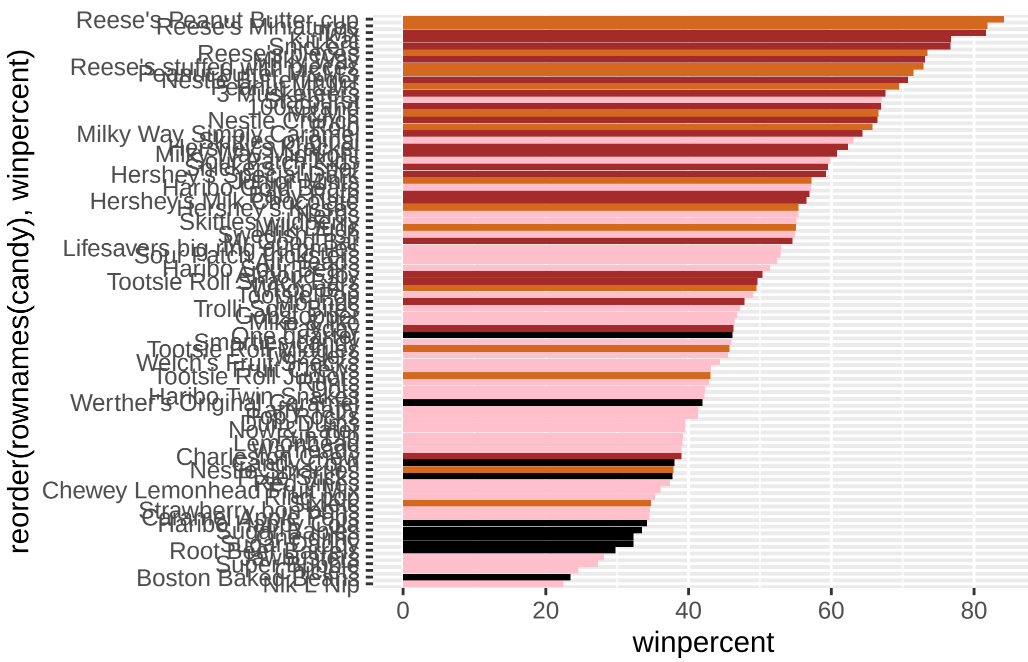
```
ggplot(candy) +
  aes(x = winpercent, y = reorder(rownames(candy), winpercent)) +
  geom_col() +
  labs(title = "Candy Rankings by Win Percentage",
       x = "Win Percentage",
       y = "Candy Type")
```



Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Boston Baked Beans.

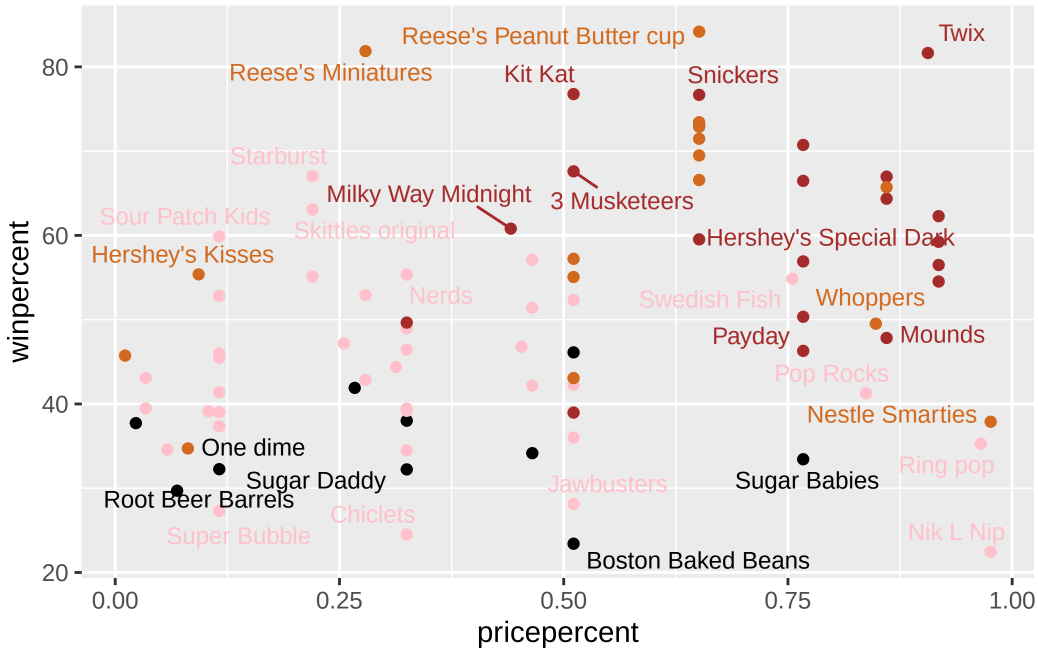
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Taking a Look at PricePercent

```r
library(ggrepel)

ggplot(candy) +
  aes(pricepercent, winpercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures has a very high winpercent and a low pricepercent.

```
candy[order(candy$pricepercent, -candy$winpercent), c("pricepercent", "winpercent")]
```

```
                          pricepercent winpercent
Tootsie Roll Midgies             0.011   45.73675
Pixie Sticks                     0.023   37.72234
Fruit Chews                      0.034   43.08892
Dum Dums                         0.034   39.46056
Strawberry bon bons              0.058   34.57899
Root Beer Barrels                0.069   29.70369
Sixlets                          0.081   34.72200
Hershey's Kisses                 0.093   55.37545
Lemonhead                        0.104   39.14106
Sour Patch Kids                  0.116   59.86400
Sour Patch Tricksters            0.116   52.82595
Smarties candy                   0.116   45.99583
Twizzlers                        0.116   45.46628
Laffy Taffy                      0.116   41.38956
Warheads                         0.116   39.01190
```

Red vines	0.116	37.34852
One dime	0.116	32.26109
Super Bubble	0.116	27.30386
Starburst	0.220	67.03763
Skittles original	0.220	63.08514
Skittles wildberry	0.220	55.10370
Trolli Sour Bites	0.255	47.17323
Werther's Original Caramel	0.267	41.90431
Reese's Miniatures	0.279	81.86626
Lifesavers big ring gummies	0.279	52.91139
Runts	0.279	42.84914
Welch's Fruit Snacks	0.313	44.37552
Nerds	0.325	55.35405
Tootsie Roll Snack Bars	0.325	49.65350
Tootsie Pop	0.325	48.98265
Mike & Ike	0.325	46.41172
Now & Later	0.325	39.44680
Fun Dip	0.325	39.18550
Candy Corn	0.325	38.01096
Caramel Apple Pops	0.325	34.51768
Sugar Daddy	0.325	32.23100
Chiclets	0.325	24.52499
Milky Way Midnight	0.441	60.80070
Gobstopper	0.453	46.78335
Haribo Gold Bears	0.465	57.11974
Haribo Sour Bears	0.465	51.41243
Haribo Twin Snakes	0.465	42.17877
Haribo Happy Cola	0.465	34.15896
Kit Kat	0.511	76.76860
3 Musketeers	0.511	67.60294
Junior Mints	0.511	57.21925
Milk Duds	0.511	55.06407
Air Heads	0.511	52.34146
One quarter	0.511	46.11650
Tootsie Roll Juniors	0.511	43.06890
Dots	0.511	42.27208
Charleston Chew	0.511	38.97504
Chewey Lemonhead Fruit Mix	0.511	36.01763
Jawbusters	0.511	28.12744
Boston Baked Beans	0.511	23.41782
Reese's Peanut Butter cup	0.651	84.18029
Snickers	0.651	76.67378
Reese's pieces	0.651	73.43499

```
Milky Way                          0.651   73.09956
Reese's stuffed with pieces        0.651   72.88790
Peanut butter M&M's                0.651   71.46505
Peanut M&Ms                        0.651   69.48379
M&M's                              0.651   66.57458
Snickers Crisper                   0.651   59.52925
Swedish Fish                       0.755   54.86111
Nestle Butterfinger                0.767   70.73564
Nestle Crunch                      0.767   66.47068
Baby Ruth                          0.767   56.91455
Almond Joy                         0.767   50.34755
Payday                             0.767   46.29660
Sugar Babies                       0.767   33.43755
Pop Rocks                          0.837   41.26551
Whoppers                           0.848   49.52411
100 Grand                          0.860   66.97173
Rolo                               0.860   65.71629
Milky Way Simply Caramel           0.860   64.35334
Mounds                             0.860   47.82975
Twix                               0.906   81.64291
Hershey's Krackel                  0.918   62.28448
Hershey's Special Dark             0.918   59.23612
Hershey's Milk Chocolate           0.918   56.49050
Mr Good Bar                        0.918   54.52645
Ring pop                           0.965   35.29076
Nestle Smarties                    0.976   37.88719
Nik L Nip                          0.976   22.44534
```

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, Hershey's Milk Chocolate. Of those candies, Nik L Nip is the least popular, since it has the lowest winpercent.

```r
ord <- order(candy$pricepercent, decreasing = TRUE)
head(candy[ord, c("pricepercent", "winpercent")], n = 5)
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
```

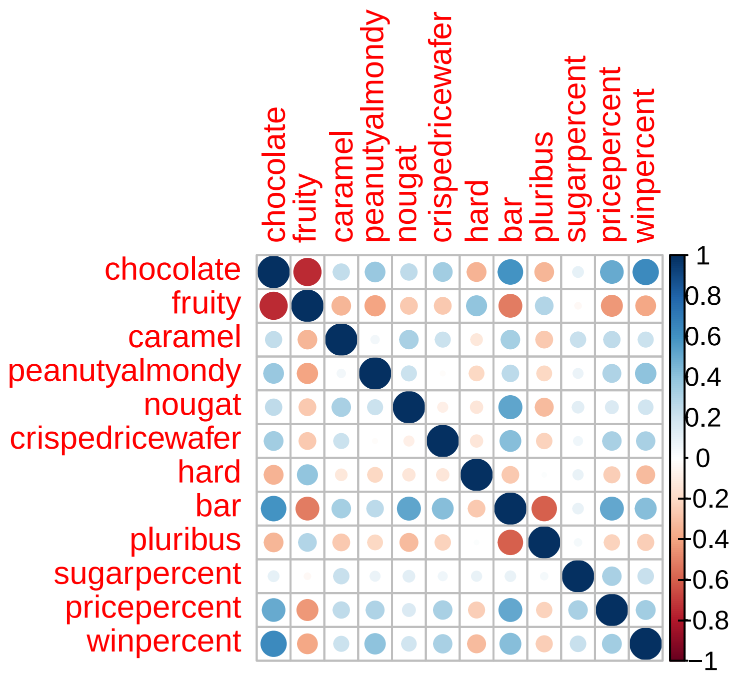
```
Hershey's Milk Chocolate        0.918   56.49050
```

Exploring the Correlation Structure

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are anti-correlated.

> Q23. Similarly, what two variables are most positively correlated?

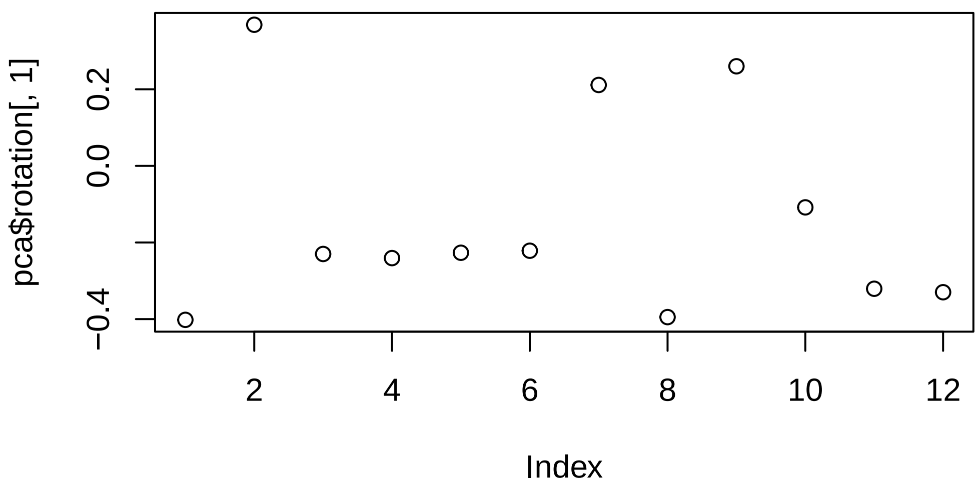
Chocolate and bar are the most positively correlated.

Principal Component Analysis (PCA)

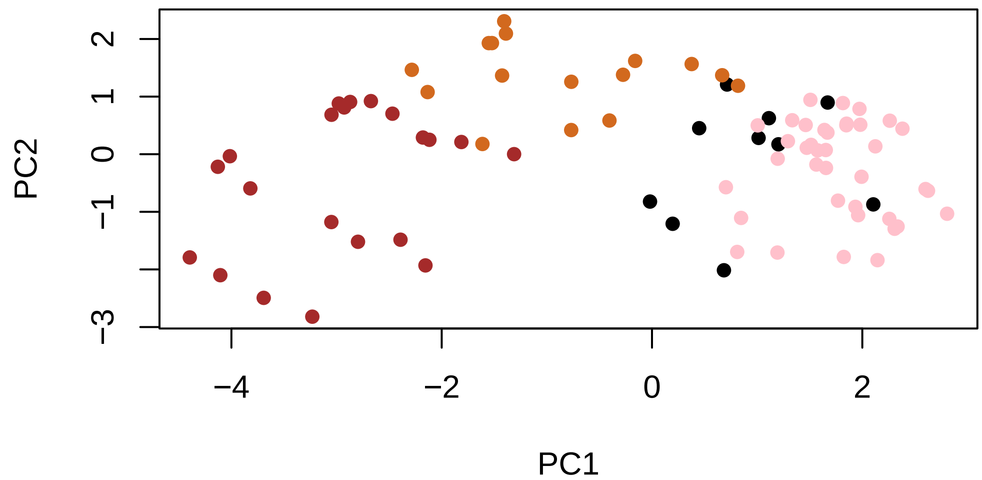
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

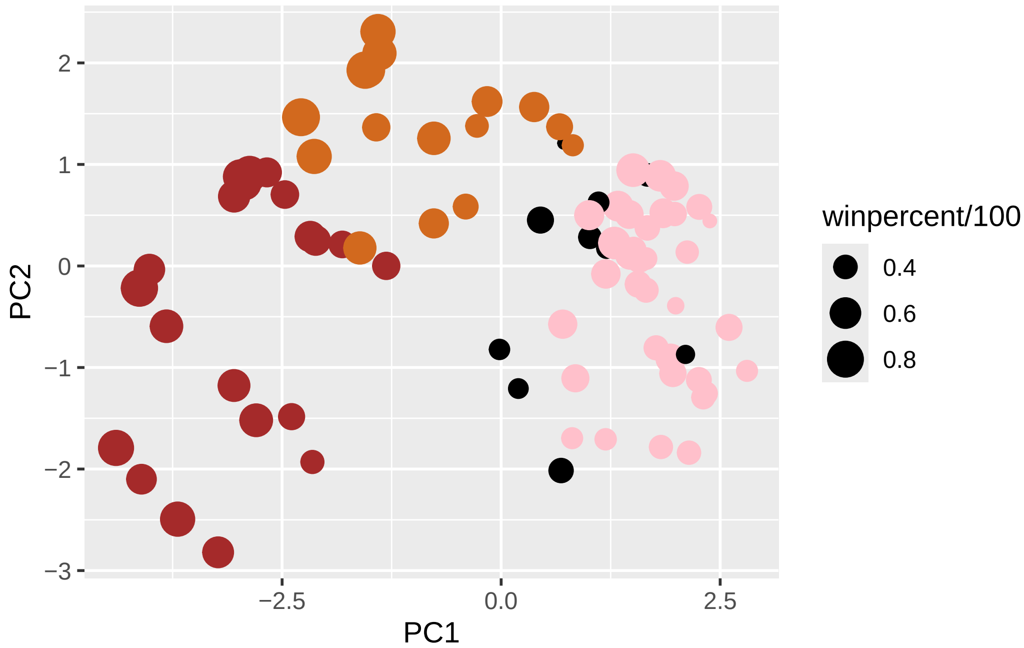
```
plot(pca$rotation[, 1])
```



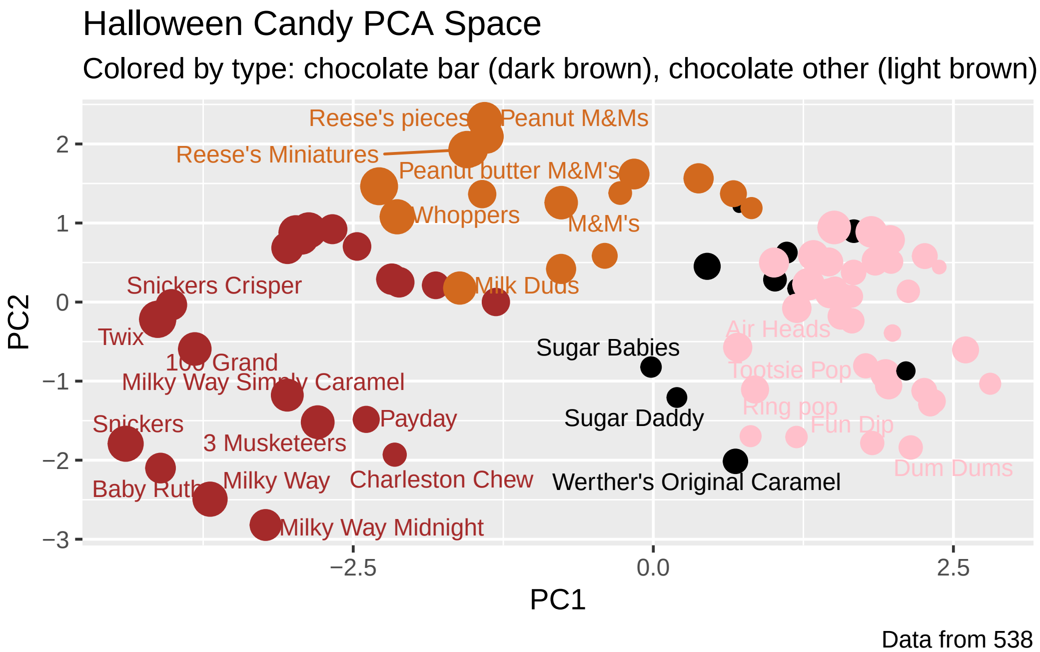

```
plot(pca$x[, 1:2], col = my_cols, pch = 16)
```




```
my_data <- cbind(candy, pca$x[, 1:3])
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent / 100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
p
```



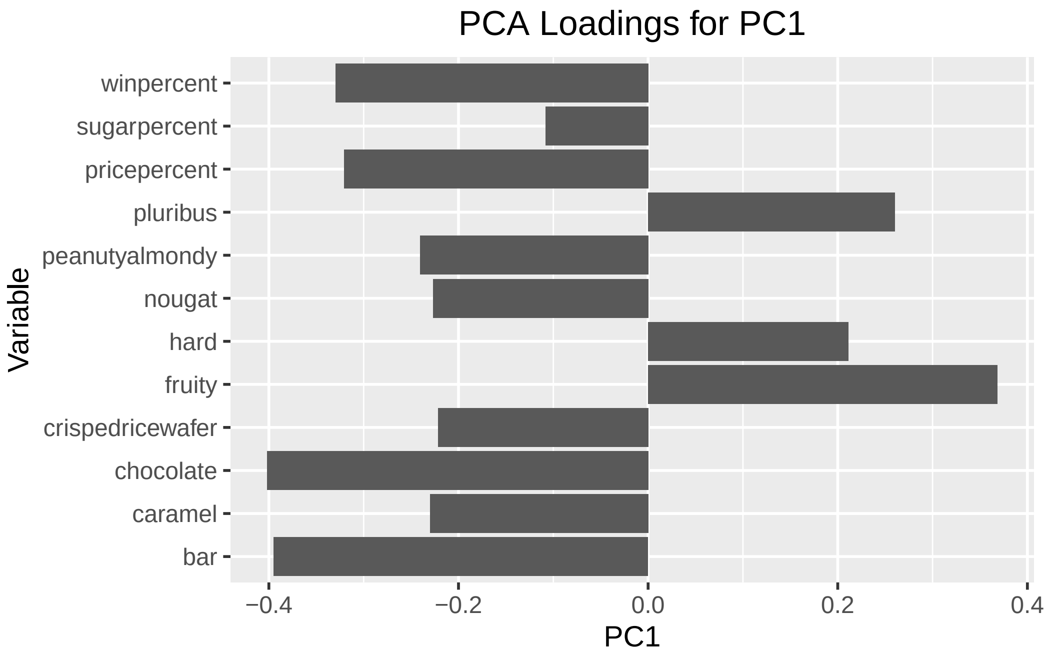

```r
library(ggrepel)
p + geom_text_repel(size = 3.3, col = my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title = "Halloween Candy PCA Space",
       subtitle = "Colored by type: chocolate bar (dark brown), chocolate other (light brown)
       caption = "Data from 538")
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The original variables that were picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. These do make sense to me because fruity candy most of the time don't have chocolate and they come in multiple pieces or harder candy. I saw this relationship highlighted in the previous correlation plot where fruity candy was negatively correlated with chocolate.

```
ggplot(pca$rotation) +
  aes(PC1, rownames(pca$rotation)) +
  geom_col() +
  labs(title = "PCA Loadings for PC1",
       x = "PC1",
       y = "Variable") +
  theme(plot.title = element_text(hjust=0.5))
```



Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The combination of characteristics that make a "winning" candy appear to be chocolate, in a bar form, and not fruity/hard. The chocolate candies have a higher winpercent compared to fruity candies. The t-test also showed the difference was statistically significant. The top ranked candies were also chocolate candy, like Reese's, KitKat, and Snickers with filling. The plots were able to help us visualize the ranking of candies and the comparison of winpercent to other variables, like pricepercent. The correlation plot showed that chocolate/winpercent are positively related and fruit/chocolate are negatively related. The PCA separated the chocolate/bar candies from the fruity/pluribus/hard candies along PC1. All of these different analyses support the conclusion that chocolate candies with filling appear to make a "winning" candy.